

EDUCATION

Courant Institute of Mathematical Sciences, New York University , computer science <i>PhD</i> .	starting 2026.9
• Verifiable systems and theoretical computer science.	
Stanford University , computer science <i>M.S.</i> and mathematics <i>B.A.</i>	2019.9 - 2024.6
• General Yaowu Wang Scholarship.	
St Paul’s School	2017.9 - 2019.6
• Full scholarship.	

PAPERS

- *The Trace-Zero Subgroup and the Relative Trace for BN254*,
Julius Zhang,
Research Note, 2026.
[pdf](#)
- *Trace Pairing*,
Julius Zhang,
Manuscript, 2026.
[pdf](#)
- *Complex Cobordism and Formal Group Law*,
Julius Zhang,
Manuscript (Undergraduate Honours Thesis), Stanford University, 2024.
[pdf](#)

EXPERIENCE

a16z	Researcher, 2025.4 - Present
• Design and build Jolt , an open-source zero-knowledge virtual-machine in Rust.	
• Proof compression research : designed torus compression system and proved theoretical optimality via Hilbert 90.	
• Audited elliptic-curve optimizations, formalizing folklore cryptographic results into rigorous proofs .	
• Performance engineering across the proving stack, including sum-check optimization.	
Millennium	Researcher, 2024.6 - 2025.4
• Convex optimization and robust statistical estimation for real-time trading systems.	
Stellar Development Foundation	Consultant, 2023.6 - 2023.9
• Byzantine fault tolerance, C++, stellar-core .	
• Performance optimization: improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.	
Distributed Systems and TCS Research, Stanford University	Researcher, 2023.9 - 2024.6
• Under David Mazieres, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules.	
• Research PCP under Aviad Rubinstein, presented full proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.	
Topology Research, Stanford University	Researcher, 2020.7 - 2022.5
• Honours thesis related to Floer homotopy theory under Mohammed Abouzaid (see Papers). Proved a result sketched by Daniel Quillen, Fields medalist, relating the Gysin homomorphism of complex cobordism to the Grothendieck residue symbol. Produced a recursive formula to compute quantum product correction term in Gromov-Witten invariant valued cohomology.	
• Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.	

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, Floer theory, gauge theory, concentration inequalities.